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Original Article

Relationship between Pre-Training Subjective Wellness Measures, Player Load and Rating of Perceived Exertion Training Load in American College Football

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ABSTRACT

Purpose: The relationship between pre-training subjective wellness, external and internal training load in American College football is unclear. This study examined the relationship between pre-training subjective wellness (sleep quality, muscle soreness, energy, wellness Z score) on 1) player load and 2) session rating of perceived exertion (s-RPE-TL) in American College footballers. **Methods:** Subjective wellness (measured using 5-point, Likert scale questionnaires); external load (derived from global position systems [GPS] and accelerometry) and s-RPE-TL were collected during three typical training sessions per week for the second half of an American collegiate football season (eight weeks). The relationship between pre-training subjective wellness and 1) player load and 2) s-RPE training load were analysed using linear mixed models with a random intercept for athlete and a random slope for training session. Standardised mean differences (SMD) denote the effect magnitude. **Results:** A one unit increase in wellness Z score and energy were associated with a trivial 2.3% (90% confidence interval (CI): 0.5, 4.2; SMD: 0.12) and 2.6% (90% CI: 0.1, 5.2; SMD: 0.13) increase in player load. A one unit increase in muscle soreness (players felt less sore) corresponded to a trivial 4.4% (90% CI: -8.4, -0.3; SMD: -0.05) decrease in s-RPE training load. **Conclusion:** Measuring pre-training subjective wellness may provide information about players' capacity to perform within a training session and could be a key determinant of their response to the imposed training demands American College football. Hence, monitoring subjective wellness may assist the individualisation of training prescription in American College footballers.

Keywords: Fatigue, sleep, recovery, monitoring, GPS.

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INTRODUCTION

Daily monitoring of a player's internal and external training loads are critical in American College football since a high training load coupled with inadequate recovery can result in injury, illness or overtraining (1). One commonly used, non-invasive method of monitoring an athlete's psychobiological training load is the session rating of perceived exertion training load [s-RPE training load: session duration (in minutes) × RPE (using either CR-10, CR-100 or 6-20 scales)] (2). Several early studies have established the construct validity of s-RPE training load against other forms of internal load (such as heart rate and blood lactate) and external load measures derived from microtechnologies such as global positioning systems (GPS) and accelerometers (3-5). Consequently, s-RPE training load is used extensively alongside GPS-derived metrics of training load (such as player load and total distance run) in football codes to monitor changes in players' training and match performance throughout a season.

In addition to monitoring external load via GPS and accelerometers, monitoring subjective ratings of wellness and mood states before each training session may provide information about a player's psychological response to the global training load in team sports (6). For example, pre-training wellness questionnaires are considered valid and reliable tools to imply changes in mood states in athletes (7) despite their unclear relationship with s-RPE training load in a team sport context. Recently, Gallo et al. (8) investigated the relationship between pre-training subjective wellness (sleep quality, fatigue, stress, mood and muscle soreness) and external load in Australian Footballers. These authors found that a one unit decrease in wellness Z score resulted in a 4.9% (95% confidence interval (CI): ± 3.1) and 8.6% (95% CI: ± 3.9) decrease in player load and player load slow (running activity < 2 $m.s^{-1}$), respectively. In essence, these results suggest lower pre-training subjective wellness scores may precede a decrease in external load during a training session, indicating that training

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prescription parameters may need to be altered to reduce an athlete's risk of injury, illness or fatigue. Although the relationship between pre-training subjective wellness, external load and s-RPE training load is established in other football codes, there is currently a limited understanding of these relationships in American College football. Further research is therefore required to describe such relationships between these training load parameters in American College football owing to the differing physical and psychological demands, playing positions, anthropometric characteristics (body mass may vary from 70-150 kg) and training/match commitments compared with other team sports (9).

To this end, this study investigated the relationship between pre-training subjective wellness measures (soreness, sleep, energy, wellness Z score) and 1) external load (player load) and 2) s-RPE training load in American College footballers.

METHODS

Participants

Fifty-eight American College footballers participated in this study [mean ± standard deviation (SD); age: 20.1 ± 1.1 y, mass: 103.9 ± 19.3 kg, height: 188.7 ± 7.0 cm]. All players were members of the same Division I National Collegiate Athletic Association (NCAA) football team. Data collection was implemented as part of the institution's athletic department performance procedures. Players provided written informed consent indicating that de-identified, wellness or performance data may be used for research. The University's Research Compliance Services approved all experimental procedures for this retrospective analysis.

Study Design

This study retrospectively analysed pre-training subjective wellness and training load data from training sessions performed during the 2015 NCAA football season. Specifically, data collation occurred for the second half of the NCAA Division 1 football season (eight-

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weeks). Each training week included five field-training sessions and four weight-training sessions. Measures herein pertain only to field-based training measures and wellness data. Training load measures included microtechonology (derived from GPS and accelerometry) and internal load (s-RPE training load), whilst subjective wellness (perceived muscle soreness, sleep, energy) were also collected. Since external load was only collected on three main field-training sessions per week (excluding two walk through sessions), the following analyses include daily external load, internal load and subjective wellness data from three sessions (session 1, 2 and 3) each week.

Session 1 typically occurred two days following a game (GD+2) and consisted of predominantly low volume, moderate intensity exercise. Session 2 occurred three days after the game (GD+3) and consisted of moderate volume and intensity. Finally, session 3 occurred four days following a game (GD+4; or three days before the next game) and was the main session of the week, focussing on high volume and intensity training. Only team training sessions were included; individual, rehabilitation and recovery sessions were excluded due to their differing modes and intensities.

Data Collection

Subjective wellness

Players rated three questionnaire items (muscle soreness, sleep and energy) on 1-5 Likert scale each morning ~2 h before field training commenced. These were collected individually via players inputting into a desktop computer database within a private area in the weight room. All scales were anchored on a 1-5 scale. The soreness scale asked, *"How SORE were you when you woke up this morning?"* (1 = terribly sore, 5 = no soreness at all). The sleep scale asked: *"How did you SLEEP last night?"* (1 = terrible sleep, 5 = excellent sleep); the energy scale asked: *"How ENERGISED do you feel today?"* (1 = no energy at all, 5 = totally

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energised). The players were familiarised with all scales two weeks before the commencement of the study. In addition, a wellness Z score was calculated for each player using the following formula: player's session subjective wellness score – player's mean subjective wellness score/standard deviation (SD) of players' subjective wellness score (8).

Internal load

Internal load was determined using the s-RPE method. Players were asked *"how physically exerting did you find the training session?"* ~30 minutes after each training session and rated their response on Borg's CR-10 scale (10), with 0 = *rest* and 10 = *maximal*. Session RPE was then multiplied by session duration (in minutes) to calculate s-RPE training load (2). Players were familiarised with the scale two weeks before the commencement of the study.

External load

Players wore a GPS unit during training and match activities (Optimeye S5; Catapult Innovations, Melbourne, Australia). The Optimeye device includes a 10 Hz GPS, a 100 Hz accelerometer and a 100 Hz gyroscope, which have previously been shown to have acceptable reliability and validity during team-sport activity (11, 12). Devices were inserted into a custom-made pouch and attached between the scapulae of the players' shoulder pads. Each player used the same GPS device each day to maintain consistency between sessions (13). Sessions were coded for individual periods but ran for the full duration of the session without omission of time. Data were uploaded post-session using Catapult's OpenField 1.11 software (Catapult Innovations, Melbourne, Australia) and collated into Microsoft Excel. Player load was calculated for each training session using a customised algorithm within the software provided by the manufacturers (OpenField 1.11 software, Catapult Innovations, Melbourne, Australia). Briefly, this parameter is collected through tri-axial accelerometers and represents the square root of the sum of the squared instantaneous rate of change in acceleration within the three

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planes divided by 100 (Catapult Innovations, Melbourne, Australia). The accelerometers measuring player load possess high inter- and intra-device reliability and are a valid tool for assessing changes in activity and direction in team sport (12).

Statistical Analysis

Player load and s-RPE training load were log-transformed before the analysis, and back-transformed to allow the results to be expressed as a percentage change in the dependent variable for each unit change in the predictor variable. Linear mixed models were fit using the *nlme* package in the R statistics programme to examine 1) the relationship between subjective wellness (sleep, soreness, energy, wellness Z score) and player load 2) the relationship between player load and s-RPE training load and 3) the relationship between pre-training subjective wellness and s-RPE training load. All models were fit with a random intercept for athlete (to calculate the between-athlete SD) and a random slope for training session (to model a separate slope for each type of training session) using an unstructured covariance matrix. Combined models (including each subjective wellness metric both separately and collectively) were compared using the Bayesian information criteria (BIC) and the model with the lowest BIC score considered parsimonious. The imprecision of parameter estimates are expressed with 90% profile CIs. The standardised mean difference (SMD) was calculated by dividing the parameter estimate by the between-subject SD. The magnitude of the SMD was interpreted using the following qualitative descriptors: < 0.2 *trivial*, 0.2-0.6 *small*, 0.6-1.2 *moderate*, 1.2-2.0 *large*, 2.0-4.0 *very large* (14). The smallest worthwhile change (SWC) in each variable was calculated as 0.2 of the between-athlete SD (15).

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RESULTS

Relationship between subjective wellness and player load

Linear mixed model parameter estimates for the effect of subjective wellness on player load is presented in Table 1. A one unit increase in wellness Z score was associated with a trivial, 2.3% (90% CI: 0.5, 4.2) increase in player load (χ^2 (1) = 4.40, P = 0.04, BIC = 332.4, SMD: 0.12). Pre-training energy was also trivially related player load (χ^2 (1) =3.03, P = 0.08, BIC = 335.3, SMD = 0.13), with a one unit increase in energy corresponding to a 2.6% (90% CI: 0.1, 5.2) increase in player load. The SWC was 4.4% (90% CI: 2.5, 7.9). In comparison, muscle soreness (χ^2 (1) =1.81, P = 0.18, BIC = 336.8) and sleep (χ^2 (1) = 2.24, P = 0.13, BIC = 336.0) were not related to player load.

Relationship between s-RPE training load and player load

Player load was trivially related to s-RPE training load (χ^2 (1) = 137.5, P < 0.01, SMD: 0.01). Specifically, a one unit increase in player load was associated with a 0.3% (90% CI: [0.2, 0.3]) increase in s-RPE training load.

Relationship between subjective wellness and s-RPE training load

Linear mixed model parameter estimates for the effect of subjective wellness on session RPE training load is presented in Table 2. The model containing all subjective wellness variables revealed neither perceived muscle soreness (χ^2 (1) = 1.97 P = 0.16), energy (χ^2 (1) = 0.03, P = 0.86) or sleep (χ^2 (1) = 0.00, P = 0.99) were related to s-RPE training load. When modelled individually, a one unit increase in muscle soreness rating (i.e. participants perceived less muscle soreness) corresponded to a trivial, -4.4% (90% CI: -8.4, -0.3) decrease in s-RPE training load (χ^2 (1) = 3.09, P = 0.08, SMD = -0.05). Neither sleep (χ^2 (1) = 0.48, P = 0.49) nor energy (χ^2 (1) = 1.07, P = 0.30) were related to s-RPE training load. Of the three subjective wellness variables, muscle soreness showed the lowest BIC score, indicating the model

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containing muscle soreness (BIC: 956.2) was more parsimonious than energy (BIC: 957.7) and sleep respectively (BIC: 958.2).

DISCUSSION

This study investigated the relationship between pre-training subjective wellness (soreness, sleep, energy, wellness Z score) and 1) external load (i.e. player load, derived from GPS and accelerometers) and 2) internal load (i.e. s-RPE training load) in American College footballers. Pre-training wellness Z score and energy were trivially related to player load, whereas pre-training muscle soreness was trivially related to s-RPE training load. Pre-training subjective ratings of muscle soreness and sleep were not related to player load, whereas energy and sleep were not related to s-RPE-training load. Firstly, the outcomes of this study provide evidence to support the measurement of pre-exercise subjective wellness measures in addition to accelerometer and GPS-derived external load measures and s-RPE training load as important foundations of a holistic player monitoring system in American College football. Secondly, these results suggest pre-training subjective wellness ratings, such as wellness Z score and energy may influence the exercise output of American Collegiate footballers during in-season training sessions, whilst muscle soreness may influence a player's response to training, as indicated by the relationship between s-RPE training load and pre-training muscle soreness. As such, practitioners should consider an athletes' pre-training subjective wellness scores when prescribing training and/or recovery.

Some pre-exercise wellness questionnaires are valid and reliable tools that may be useful to imply changes in mood states and perceptual fatigue in athletes (7). However, there is currently no consensus on how they should be used to assist with training prescription or how they are associated with training output variables. The trivial, albeit significant relationship, between wellness Z score and player load observed in the current study suggest

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that a higher wellness Z score (i.e. the player felt better overall) was associated with a higher player load during training. Such findings support those of Gallo et al. (16), who investigated pre-training subjective wellness score (a combination of sleep quality, fatigue, stress, mood and muscle soreness) and s-RPE training load in Australian Footballers and also observed a trivial relationship between wellness Z score of −1 (d = 0.06 ± 95% CI: 0.28) and s-RPE training load. These authors speculated that players with low wellness scores might have decreased their external load to maintain their RPE during a training session. In addition, Crowcroft et al. (17) concluded that general health was a more sensitive diagnostic tool to measure performance changes in National-level swimmers than any individual pre-training wellness variable (i.e. soreness, motivation, total quality recovery and fatigue) alone. Hence, averaging a player's pre-training subjective wellness measured across several wellness variable may yield superior information about their potential training performance (as indicated by their player load) than any single wellness variable measured in isolation. By contrast, one limitation of using average measures is a loss of sensitivity, although such a loss in information can be overcome by calculating a correct weighting factor for each wellness measure in relation to a global wellness state. Furthermore, the development of a "wellness passport", based on the adaptive Bayesian network approach currently utilised in athlete biological passport (18) could help to integrate information from both subjective wellness and external load measures to estimate global wellness. Such a model may also be useful in providing dynamic, individualised reference ranges for both internal and external load measures, consequently improving our ability to estimate an athlete's risk of injury, illness or fatigue given their historical training and match data. However, further research is necessary to establish the utility of potential objective and subjective markers and their respective importance to global wellness before being able to identify which variables should be included in a potential "wellness passport".

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Given wellness is a multidimensional construct (19), one limitation of averaging pre-training subjective wellness across several variables to represent global wellness is that it may restrict the ability to identify specific relationships between individual wellness components and different global, external and internal load variables (8). In the current study, in comparison to external load, muscle soreness, but not wellness Z score was related to s-RPE training load. Accordingly, muscle soreness may specifically contribute towards a player's response to a training stimulus (i.e. internal load measured by s-RPE training load) in American College football. Measuring pre-training muscle soreness may therefore be of particular importance in American College football to determine whether players are able to cope with both the physiological and psychological stress of training.

The relationship between pre-training ratings of muscle soreness and s-RPE training load may be explained by the impacts derived from physical contact associated with American Football. For instance, perceived muscle soreness could take longer than 4 days to return to pre-game levels in DI players (20). This suggests perceptual muscle soreness responds to short term reductions in muscle damage and power and peak force incurred from the loading demands of American Football (resulting either from training or games) (21). Indeed, Wellman and colleagues analysed the intensity, number and distribution of impact forces experienced by football players during competition and showed that wide receivers sustained more 5.0-6.5 G force impacts (moderate-to-light) than other position groups, whereas running backs endured the most severe (>10 G force) impacts (other than the quarter backs) (9). Muscle tissue damage following game impacts may therefore explain the relationship of muscle soreness and s-RPE training load observed in the present study. However, many studies to date have examined changes in player wellness and/or external load between games, but additional training-based investigations of the relationship between these variables in American College football are necessary to confirm our findings. Moreover, several variables affect s-RPE training load, such

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as playing experience, position and markers of fitness (22). Future studies analysing the relationship between pre-exercise subjective wellness measures and s-RPE training load should account for these factors and assess the how the magnitude of the relationship changes between different periods of the season, where accumulated fatigue may uncouple the relationship between pre-exercise subjective wellness measures and s-RPE training load.

We chose player load rather than total distance as an explanatory variable since we believe accelerometer-based metrics (such as player load) may better represent the physical demands of American Football (9). For instance, many positions within American Football, such as the trench positions, cover very short total distances and perform a high amount of impacts, collisions accelerations and decelerations that are captured by the player load metric. s-RPE training load is reportedly highly correlated with total distance run (r = 0.80 [0.72, 0.86]) and player load (r = 0.84, [0.77-0.89]) in soccer players (23). In contrast to Scott et al. (23) we observed a trivial, albeit significant, relationship between s-RPE-TL and player load in the current study. Such differences in the relationship between s-RPE-TL and player load may result from the different match demands of American College football and soccer and the different statistical modelling procedures (i.e. random slopes and intercepts model used in the current study compared with ordinary least squares regression procedures elsewhere). For example, Bartlett et al. (24) analysed s-RPE training load data from a training season in Australian Rules Footballers using a neural network and an individualised GEE analysis finding that total distance exhibited the best relationship with s-RPE training load, rather than high speed running and player load. Similarly, a possible difference between Bartlett et al. (24) and the current study may be the manner in which the player load variable is accumulated in Australian Football *versus* American football. For instance, Cormack and colleagues (25) reported that in Australian Football, fatigued players had a lower contribution of vertical acceleration to player load, yet were able to maintain high-speed running and total distance

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over the course of a game. Since the player load variable has been suggested to incorporate changes in speed, direction and impacts (26), it is possible that the player load variable is more representative of the nature of NCAA football where less distance and high-speed running is performed compared to Australian Football. Indeed, NCAA football is comprised of short periods of explosive plays with long rest periods between plays (to accommodate for television advertising). As a result, during NCAA football the demands may be as high as 38 high acceleration efforts and impact forces of up to 10 G-force units, along with total distances of 5,530 m and 655 m of high intensity running respectively (9). These characteristics of collegiate football may contribute towards the association between player load and s-RPE training load in the current study. Hence, in the absence of microtechnology, s-RPE training load may provide a useful measure of external training load in American College football.

Limitations

This manuscript has several limitations. The subjective wellness questionnaire used in this study reflects custom measures widely used in practice, however, they have not undergone a rigorous process of development and evaluation to ensure responses have an acceptable degree of validity and reliability.. Secondly, some training information was unsuitable for analysis since these sessions were conducted indoors (thus removing the ability to collect GPS measures). Finally, data collection was conducted in the middle and the end of the regular season. More research is therefore required to confirm whether our findings are valid at the beginning of the season and whether such results are consistent across multiple seasons.

PRACTICAL APPLICATION

Collectively, our findings support the measurement of pre-training subjective wellness measures in addition to GPS-derived external and s-RPE training load as important foundations of a player monitoring system in American College football. These data are important for

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practitioners since they show that perceived pre-training muscle soreness may be a key determinant of a player's s-RPE training load in American College football players. Furthermore, longitudinal monitoring of an athlete's s-RPE training load in tandem with pre-training subjective wellness measures provide a simple, global measure of an athlete's wellness state in American College football, which may assist coaches and sport scientists to more accurately anticipate a student athlete's risk of injury or illness. Finally, since pre-training perceived muscle soreness was associated with fluctuations in s-RPE training load, developing effective methods to recover from muscle soreness and to accurately quantify the training stress resulting from collisions may help to improve the monitoring of global wellness in American College football.

CONCLUSION

This study investigated the relationship between external load measures and pre-training subjective wellness on s-RPE training load in American College footballers. Subjective wellness Z score and energy were related to player load indicating that pre-training wellness state may partially determine performance in training. Additionally, perceived muscle soreness was related to s-RPE training load, perhaps highlighting that muscle soreness is a key contributor of a player's response to the imposed training demands. The outcomes of this study could be used to provide evidence supporting the measurement of pre-exercise subjective wellness measures in addition to GPS-derived external load and s-RPE training load as important foundations of an holistic player monitoring system in American College football.

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Table 1: Linear mixed model parameter estimates and 90% confidence intervals (CI) for the relationship between external load (dependent variable - player load) and internal (session rating of perceived exertion training load) and pre-training subjective wellness measures (soreness, energy, sleep, wellness Z score).

	Model 1		**Model 2**		**Model 3**		**Model 4**		**Model 5**		**Model 6**	
Fixed Effects	Est.	90% CI	Est.	90% CI	Est.	90% CI	Est.	90% CI	Est.	90% CI	Est.	90% CI
Intercept	141.2	[122.8, 162.3]	288.0	[265.8, 311.9]	280.7	[256.5, 307.2]	283.8	[258.9, 311.1]	302.4	[290.4, 314.9]	273.1	[245.7, 303.5]
s-RPE-TL	0.3	[0.2, 0.3]										
Soreness			1.9	[-0.4, 4.2]							0.9	[3.5, 146]
Energy					2.6	[0.1, 5.2]					1.1	[4.2, 200.4]
Sleep							2.3	[-0.2, 4.9]			1.6	[4.9, 395.3]
Wellness									2.3	[0.5, 4.2]		
SWC (%)												
S1	10.8	[7.1, 16.9]	4.3	[2.5, 7.7]	4.4	[2.5, 7.9]	4.4	[2.5, 7.8]	4.3	[2.5, 7.6]	4.4	[2.3, 8.9]
S2	22.9	[17.6, 30.3]	8.6	[6.8, 10.9]	8.7	[6.9, 11.0]	8.7	[6.9, 11.0]	8.9	[7.0, 11.2]	8.8	[6.7, 11.7]
S3	23.5	[17.9, 31.4]	10.1	[8.0, 12.7]	10.2	[8.1, 12.9]	10.1	[8.1, 12.8]	10.2	[8.1, 12.9]	10.3	[7.8, 13.6]
Model Fit												
BIC	963.57		336.80		335.32		336.04		332.37		360.25	

Abbreviations: s-RPE-TL = Session rating of perceived exertion training load; SWC = Smallest worthwhile change; S1 = session 1; S2 = session 2; S3 = session 3, BIC = Bayesian information criteria.

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Table 2: Linear mixed model parameter estimates and 90% confidence intervals (CI) for the relationship between internal load (dependent variable - session rating of perceived exertion training load) and external load (player load) and pre-training subjective wellness (soreness, energy, sleep, wellness Z score) measures.

	Model 1		**Model 2**		**Model 3**		**Model 4**		**Model 5**		**Model 6**	
Fixed Effects	Est.	90% CI	Est.	90% CI	Est.	90% CI	Est.	90% CI	Est.	90% CI	Est.	90% CI
Intercept	141.2	[122.8, 162.3]	608.5	[532.3, 695.7]	589.5	[508.6, 683.3]	573.1	[490.2, 669.9]	558.7	[526.8, 592.6]	615.0	[515.2, 734.0]
Player Load	0.3	[0.2, 0.3]										
Soreness			-4.4	[-8.4, -0.3]							-4.1	[-8.8, 0.7]
Energy					-2.7	[-6.9, 1.7]					-0.6	[-6.3, 5.4]
Sleep							-2.0	[-6.5, 2.8]			0.0	[-5.6, 6.0]
Wellness									-0.1	[-3.5, 3.5]		
SWC (%)												
S1	10.8	[7.1, 16.9]	31.5	[19.7, 53.6]	32.40	[20.1, 55.9]	32.4	[21.0, 52.7]	35.2	[22.9, 57.1]	31.5	[21.3, 48.6]
S2	22.9	[17.6, 30.3]	34.4	[24.6, 49.8]	34.60	[24.4, 50.7]	34.8	[26.2, 47.4]	35.1	[26.0, 48.7]	34.4	[26.2, 46.2]
S3	23.5	[17.9, 31.4]	41.5	[29.3, 60.8]	41.50	[29.0, 61.8]	41.7	[31.2, 57.4]	42.3	[31.0, 59.4]	41.4	[31.2, 56.3]
Model Fit												
BIC	963.57		956.21		957.72		958.18		960.62		978.82	

Abbreviations: SWC = Smallest worthwhile change; S1 = session 1; S2 = session 2; S3 = session 3, BIC = Bayesian information criteria.